Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/21/2005
PI Name:	Juergensmeyer, Margaret A Ph.D.		
Project Title:	Gene Expression Profiles for E. coli in Common Modeled Microgravity Systems		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60616	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	03/01/2004	**End Date:**	05/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	2/07--Received NCE to 5/31/2007, per A. Chu (ARC). 8/06--funding reinstated thru 2/28/07, per J. Cohen (HQ)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Juergensmeyer, Elizabeth (Judson College) Mobley, Evelyn (IIT Research Institute)		
Grant/Contract No.:	NNA04CC75G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bacteria are known to respond to the spaceflight environment. Literature reviews reveal a wide array of responses, from antibiotic resistance to changes in growth curve kinetics. However, because it is extremely rare for an investigator to perform experiments on multiple flights, and because flight hardware is not standardized, it is often difficult to determine which element of the spaceflight environment is responsible for the observed physiological change. It is often assumed that microgravity or microgravity-induced physical changes (e.g. lack of convective movement) are responsible, but at present, there is no way of confirming this. In addition, for nearly every paper that identifies a change and attributes that change to the spaceflight environment, there is a paper that mentions no change observed. This lack of change may be even more prevalent than the literature reveals, as it is difficult to publish a paper that shows no changes in response to the treatment. Bacteriologists interested in the response of their chosen organism to the spaceflight environment often utilize a variety of modeled microgravity (MMG) systems. Slow clinostats (approx. 1/4 rpm), fast clinostats (approx. 60 rpm), and High Aspect Rotating Vessels (HARVs) have all been used as MMG systems. Previous research from this laboratory has determined that linear vibration, a facet of the spaceflight environment, can induce some of the physiological changes observed during and attributed to spaceflight itself. The primary goal of this project was to identify the changes in gene expression that occur in various types of MMG, with the future goal of comparing them with changes shown during actual space flight. We proposed to meet this primary goal by growing Escherichia coli K-12, a common research organism that has often flown on orbital platforms, in a wide array of MMG systems. Growth curves would be performed in each MMG, and bacterial samples would be collected at equivalent points in the growth curve. Total RNA would be isolated from the samples and analyzed using Affymetrix gene chip microarrays, which would identify changes in gene expression in each sample. A comparison across the MMGs would firstly identify genes that were changed in response to the known MMGs, thus indicating which pathways were activated in a given ground control or experimental treatment. In addition, a future flight experiment, whether done by ourselves or another researcher, can be compared to this data, allowing researchers to identify which MMG is most similar to flight results. This will allow the formation of baseline data, so that at some point in the future, flight cultures can be examined utilizing the same techniques, and a comparison of the flight cultures and the baseline data will allow identification of the optimal MMG that mimics the spaceflight environment. In addition, a comparison of MMGs will identify those that best model specific subsets of the microgravity system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In addition to ultimately enabling the identification of an optimal MMG for future ground research, this project is expected to identify gene pathways connected with motility, enhanced growth, and nutrient utilization in E. coli.
Task Progress:	Over the last year, significant progress has been made on this project. Bacterial cultures were grown and characterized in the slow clinostat, the fast clinostat, and under linear vibration. Mid-log-phase samples were collected from each MMG, and pooled to achieve optimal volume. Total RNA was isolated from each sample using the Qiagen RNeasy kit. RNA samples of sufficient concentration and amount were tested to ensure minimal degradation and contamination with DNA, and sent to the University of Chicago for chip hybridization and analysis. Results from this analysis indicate that fast clinorotation and vibration are very similar. Several genes showed identical regulation changes in the two MMGs, and the overall pattern of gene expression was highly similar. Slow clinorotation, however, was surprisingly different from either of the other two MMGs. A large number of genes were up-regulated in this MMG, to an extent that the cutoff point for significance had to be increased to avoid being overwhelmed with data. Even so, many of the changes in regulation corresponded with the other two MMG systems, indicating that similar pathways respond to these stressors. These results were presented in two posters at the 2005 American Society for Gravitational and Space Biology meeting. The original intent was to present one poster at the American Society of Microbiology meeting in May of 2005, and the second at the November ASGSB meeting but, due to illness, both posters were presented at the later meeting.
Bibliography Type:	Description: (Last Updated: 09/04/2007)
Abstracts for Journals and Proceedings	Juergensmeyer EA, Juergensmeyer MA, Nelson ES. "Vibration Without Oxygenation Affects Bacterial Growth." American Society for Gravitational and Space Biology annual meeting, Reno, NV, November 2005. Gravitational and Space Biology 2005 Nov;19(1):36. , Nov-2005
Abstracts for Journals and Proceedings	Mobley EM, Juergensmeyer EA, Juergensmeyer MA. "Gene Array Analysis of E. coli Grown in Modeled Microgravity." American Society for Gravitational and Space Biology annual meeting, Reno, NV, November 2005. Gravitational and Space Biology 2005 Nov;19(1):58. , Nov-2005
Presentation	Juergensmeyer, E.A.; Juergensmeyer, M.A. and Nelson, E.S. "Vibration Without Oxygenation Affects Bacterial Growth" American Society for Gravitational and Space Biology annual meeting Nov-2005
Presentation	Mobley, E.M.; Juergensmeyer, E.A. and Juergensmeyer, M.A. "Gene Array Analysis of E. coli Grown in Modeled Microgravity" American Society for Gravitational and Space Biology annual meeting Nov-2005